FastSpeech 2 Text-to-Speech for English and Telugu

Technical Report — Implementation of FastSpeech 2-Based Text-to-Speech Synthesis
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Code: GitHub repository linked in the submission form · Checkpoints: https://huggingface.co/rohan1822/fastspeech2-tts-checkpoints

1. Implementation approach

Two separate FastSpeech 2 models (English, Telugu) were trained from the open-source implementation by ming024 (github.com/ming024/FastSpeech2, MIT licence), following Ren et al., *FastSpeech 2* (2021). Components reused unchanged: the Transformer encoder and mel decoder, the variance adaptor (duration, pitch and energy predictors with a length regulator and 256-bin linear quantised pitch/energy embeddings), the PostNet, the loss (mel, PostNet mel, pitch, energy, log-duration), the Adam optimiser with warm-up schedule, and the HiFi-GAN universal vocoder (22,050 Hz, 80 mel bins, hop 256).

Architecture. 4 encoder layers and 6 decoder layers, 2 attention heads, hidden size 256, convolution filter 1024 with kernels (9, 1), dropout 0.2; variance predictors with filter size 256, kernel 3, dropout 0.5; pitch and energy at token (phoneme/character) level, normalised; multi-speaker embedding; maximum sequence length 1500. The English model has 35.46 M parameters (115 speaker embeddings).

Modifications (shared by both languages).

- **Character-level input.** The symbol table (`text/symbols.py`) was extended with `en_a…en_z` (English) and `tel_a…tel_z` (Telugu) plus a word-space token `sp`. Training transcripts are stored as `{en_h en_e … sp …}` so the existing curly-brace path in `text/__init__.py` maps each token to a symbol id.
- **CTC forced alignment instead of Montreal Forced Aligner.** Durations come from `ctc-forced-aligner` 1.0.2 (MMS-300M ONNX; CC-BY-NC 4.0, used for alignment only). Alignments are written as TextGrids that the stock preprocessor consumes.
- **Compatibility fixes** for current libraries: librosa API in `audio/stft.py`, a NumPy-to-Torch conversion of energy in `utils/tools.py`, and PyTorch 2.6+ checkpoint loading (`weights_only=False`, `map_location`).
- **New** `inference.py` per language (the stock `synthesize.py` only handles English phonemes and Mandarin). It tokenises text, synthesizes with the 20,000-step checkpoint and HiFi-GAN, and saves audio, the predicted duration/pitch/energy arrays and the variance-adaptor plots. Pitch and energy are de-normalised with `stats.json`.
- **Split handling.** The stock preprocessor with `val_size: 0` writes every clip to `train.txt`, so train/validation lists are rewritten from our own split.

Telugu-specific. Telugu text is romanised with Uroman, lower-cased, reduced to a–z and spaces, and each letter becomes `tel_<letter>` with `sp` between words. Training tokens were reproduced exactly by this function for 200 of 200 checked training lines, and the same function is used at inference. The preprocessing config keeps `english_cleaners` because the stored text is already tokenised. **English-specific.** Text is lower-cased, non-letters removed, letters become `en_<letter>`; there is no G2P or lexicon, so the model must learn spelling-to-sound mapping itself.

2. Datasets, preprocessing and splits

2.1 English — Svarah (ai4bharat/Svarah)

3,818 clips (115 speakers, accented Indian English) were selected to target 5 h train / 1 h validation, split at clip level with every speaker present in training (no validation-only speakers). Alignment with the CTC aligner rejected clips whose first or last aligned character differed from the librosa speech boundary by more than 0.3 s: 2,779 accepted, 1,039 rejected. A recovery pass re-aligned the rejected clips with a relaxed edge check and acceptance limits calibrated on 300 accepted clips (character speaking rate, maximum character duration, speech outside the aligned span); it accepted 711 more. Recovery is a heuristic: I listened to 12 recovered clips (8 random, 4 highest-margin) and found the boundaries correct, but the rest were not manually verified. 328 clips stayed unaligned and are excluded.

2.2 Telugu — IndicTTS Telugu (SPRINGLab/IndicTTS_Telugu)

8,576 recordings (36.71 h) were filtered to 2–40 s (8,010 clips, 30.48 h) and a subset of 2,964 clips was aligned (clips longer than 12 s were skipped by the aligner); 2,913 aligned (98.3%), 51 failed (49 edge-mismatch, 2 too many letters). The dataset has two speaker labels that are gender proxies, not true speaker identities, and the split is clip-level and gender-balanced; it is **not** speaker-independent.

2.3 Final sizes

	Train clips	Val clips	Train h (mel length)	Val h (mel length)	Train h (metadata)	Val h (metadata)
English	2,920	570	4.21	0.85	4.51	0.91
Telugu	2,427	486	4.13	0.83	4.91	0.98

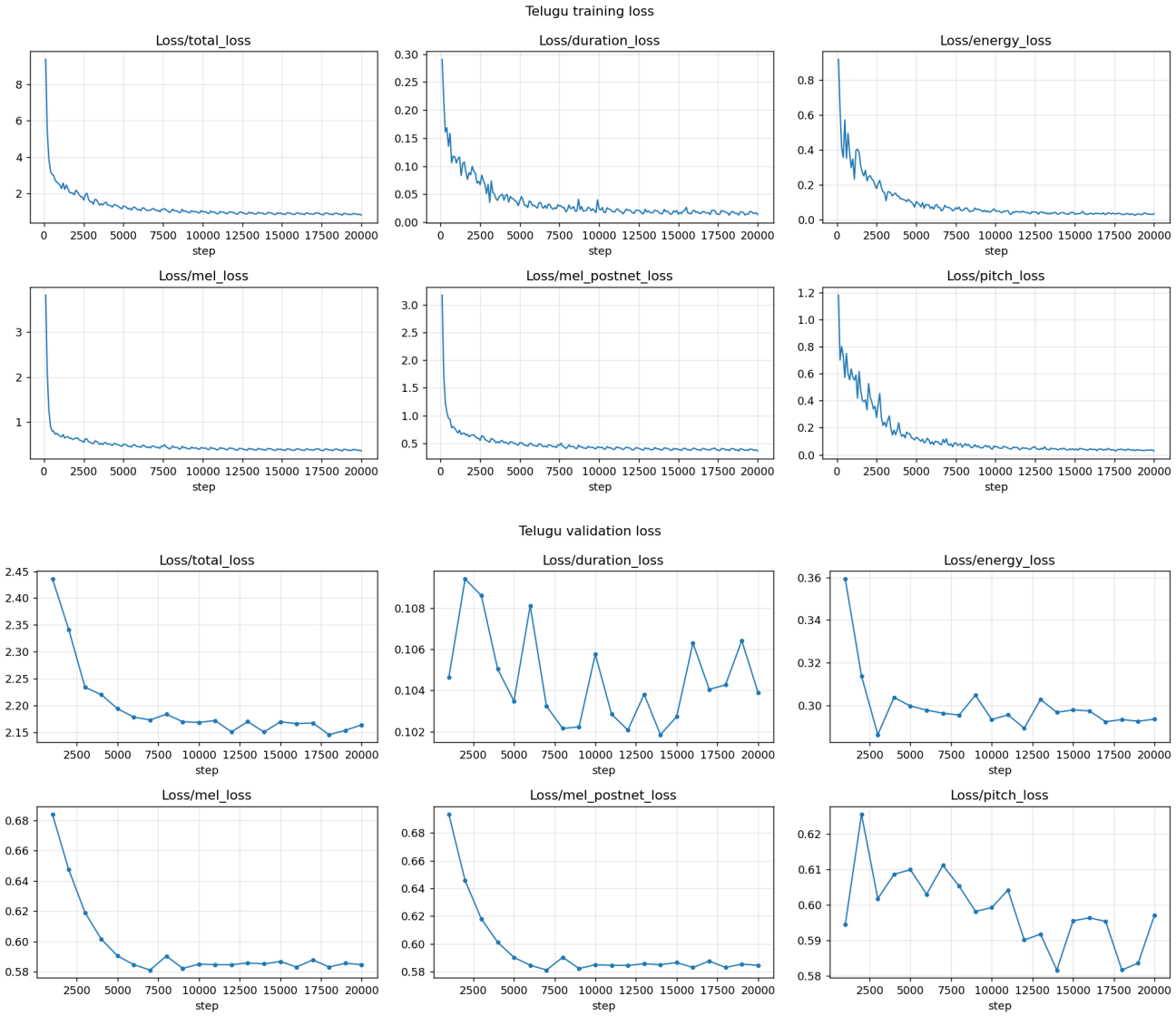
Hours are given two ways: from the extracted mel lengths (hop 256 at 22,050 Hz; the speech actually used for training) and from the source metadata durations. They differ, and I did not establish why (silence trimming at alignment is a plausible but untested explanation). Both languages are about 15% below the 5 h / 1 h target by mel length. Train and validation clips do not overlap.

3. Training configuration

Each model was trained for 20,000 steps with batch size 16, Adam (β=(0.9, 0.98), ε=1e-9), 2,000 warm-up steps, gradient clipping 1.0, validation and synthesis every 1,000 steps, checkpoints every 2,000 steps, on a Kaggle T4. Training took about 3 h 03 min (Telugu) and 2 h 23 min (English). Configs are in `config/<Language>/{preprocess,model,train}.yaml`; `config.yaml` is a combined view. Checkpoints are kept separate per language.

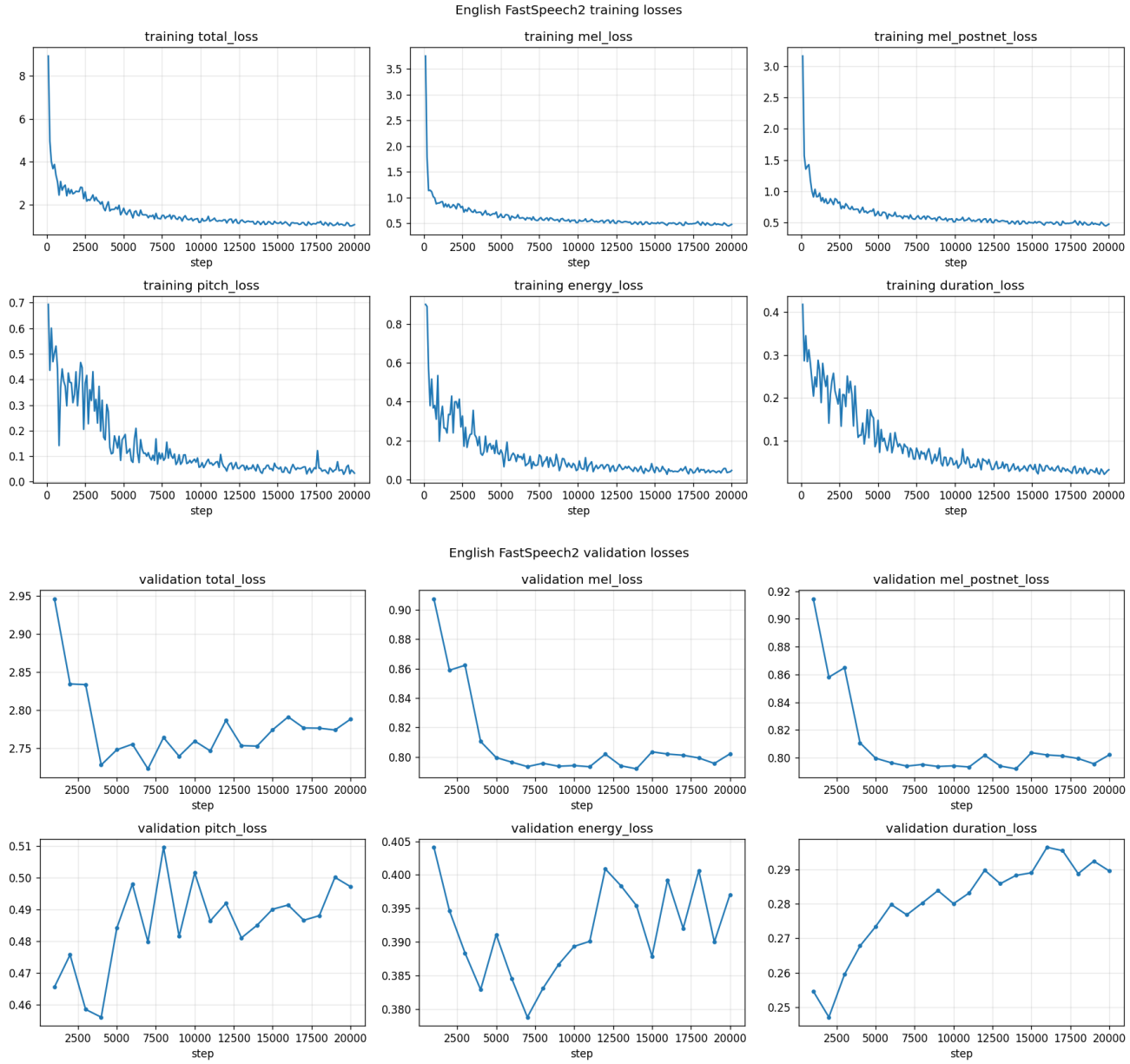
4. Training and validation loss analysis

Telugu



Validation total loss falls from 2.436 (step 1,000) to about 2.17 by step 7,000, then fluctuates between 2.146 and 2.184 with no clear trend; mel loss is flat near 0.585 from step 6,000. Best validation total was 2.1457 (step 18,000); the final value is 2.1639, a difference within the step-to-step noise. Training loss (mean of logged points) drops from 3.82 (steps 100–1,000) to 0.96 (around step 10,000) and 0.84 (final 1,000 steps). Interpretation: early convergence then a plateau, with no clear deterioration of validation loss; the last 13,000 steps added little. Convergence is not perfect, as the curve is noisy.

English



Validation mel loss falls from 0.907 to about 0.79 by step 5,000 and then stays at 0.79–0.80. The best validation total was 2.7228 (step 7,000) and the best mel loss 0.7920 (step 14,000); the final validation total is 2.788. Duration loss rises steadily after step 2,000 (0.247 to 0.290) and pitch loss drifts from about 0.46 to 0.50, a mild overfitting signal in the variance predictors. Training loss falls from about 4.0 to 1.06. The 20,000-step checkpoint was kept; it was not the best checkpoint by validation loss (checkpoints exist every 2,000 steps, so step 7,000 was never saved). A 14,000-step vs 20,000-step comparison on one test sentence was inconclusive.

Final validation (step 20,000)	Total	Mel	PostNet	Pitch	Energy	Duration
Telugu	2.1639	0.5847	0.5846	0.5971	0.2937	0.1039
English	2.7880	0.8022	0.8021	0.4971	0.3971	0.2895

Caveat. Training losses are batch averages with dropout active, while validation is computed on the full validation set, so the train/validation gap is not a pure overfitting measure. English is harder (higher mel and duration loss), plausibly because it has 115 accented speakers vs two Telugu labels, but no ablation was run to confirm this.

5. Test sentences: generated speech and variance-adaptor analysis

All sentences were synthesized with the 20,000-step checkpoints and the HiFi-GAN universal vocoder. Telugu uses speaker id 0. English uses speaker id 94 (spk028), chosen as the lowest mean WER (0.160, runner-up 0.181, range 0.160–0.255) over 20 held-out validation sentences synthesized for 12 candidate speakers and transcribed by Whisper-small. With 20 sentences per speaker this is a noisy proxy and the top candidates are within its noise; a first run of this selection had a bug and was discarded.

Text handling. Sentences are kept as given. The tokenizers strip punctuation (commas give no explicit pause; “orchestra’s” becomes “orchestras”). In E4 and T4 the digits were written out as words, because no training transcript contains digits (E4: “ten forty five a m on twelve november twenty twenty six”; T4: the year, day and time in Telugu words); the original sentence is kept alongside. E3 is synthesized as given (“rhyth m”, two words) and “andaroo” in E4 is kept. In the supplied PDF, T3 renders with broken glyphs; the intended sentence was synthesized. T1 was first typed from the PDF text and then corrected after a visual check (ఫొర్మ్); the T1 audio and plots were regenerated.

> No human listening test was performed. Quality statements for English rest on Whisper-small transcripts (an approximate proxy) and for both languages on the objective diagnostics below. Pitch and energy are token-level predictions, not frame-level F0 tracks; no token is predicted unvoiced, so the pitch plots are smooth.

5.1 English summary

ID	chars	pred. s	char ms	space %	F0 mean	F0 std	F0 drop (1st–2nd half)	energy peak
E1	98	7.27	66	10.5	191	24	+33	a@14
E2a	60	4.64	69	10.8	194	23	+15	o@4
E2b	64	4.64	67	7.0	190	27	+35	a@8
E3	42	3.04	65	10.3	196	19	+19	h@5
E4	93	6.95	63	15.2	194	20	+20	a@19

5.2 Telugu summary

ID	chars	pred. s	char ms	space %	F0 mean	F0 std	F0 drop (1st–2nd half)	energy peak
T1	87	5.18	57	4.3	214	38	+32	u@3
T2a	39	3.34	65	24.0	230	34	+9	a@2
T2b	44	2.95	60	11.0	218	40	+15	m@2
T3	37	2.67	60	17.0	212	42	+37	i@3
T4	140	11.26	59	26.9	207	26	+15	i@1

5.3 English sentences

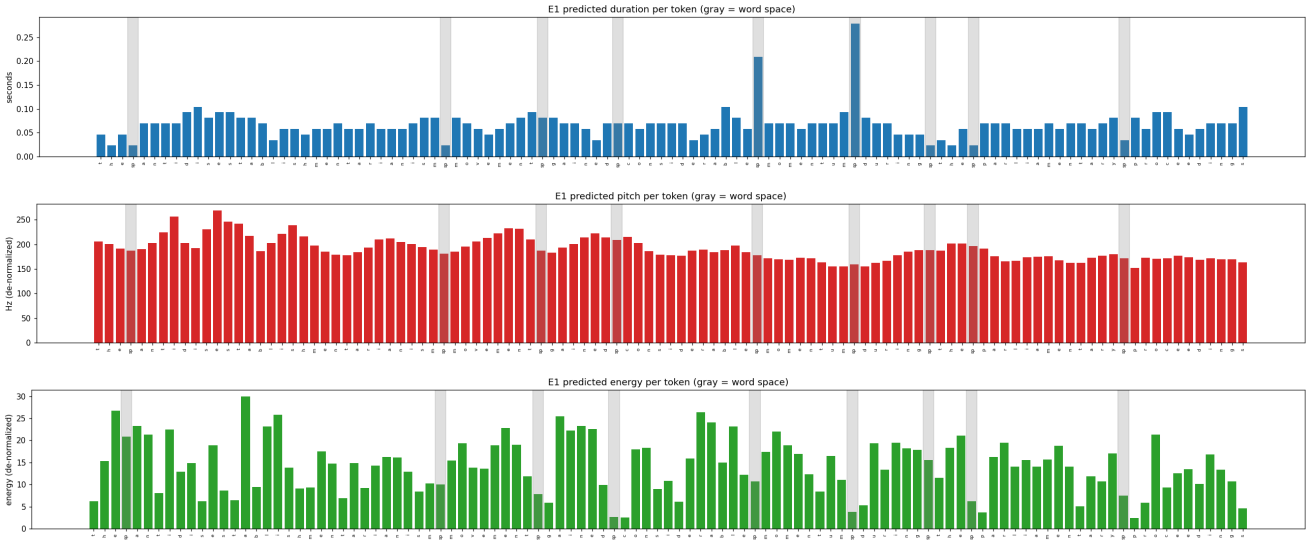
E1

The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

Speech observations. Whisper-small heard: “The anti-disease-establishmentarian is banned considerable momentum during the parliamentary proceedings.”. WER 0.30 (case, punctuation, hyphens and apostrophes ignored).

Duration. 98 character tokens in 10 words; predicted length 7.27 s (rendered audio 7.27 s). Mean character duration 66 ms (max 104 ms); word spaces average 85 ms and take 10.5% of the time. Longest tokens: 'i' (token 9, 104 ms), 'b' (token 58, 104 ms), 's' (token 106, 104 ms).
Pitch (F0). Mean 191 Hz, std 24 Hz, range 152–269 Hz. The first half of the sentence averages 208 Hz and the second half 174 Hz (a fall of 33 Hz). Highest pitch on 'e' (token 11).

Energy. Mean 14.7, range 2.4–29.9; peak on 'a' (token 14 of 107, 13% into the sentence). Duration–energy correlation across characters: -0.23.

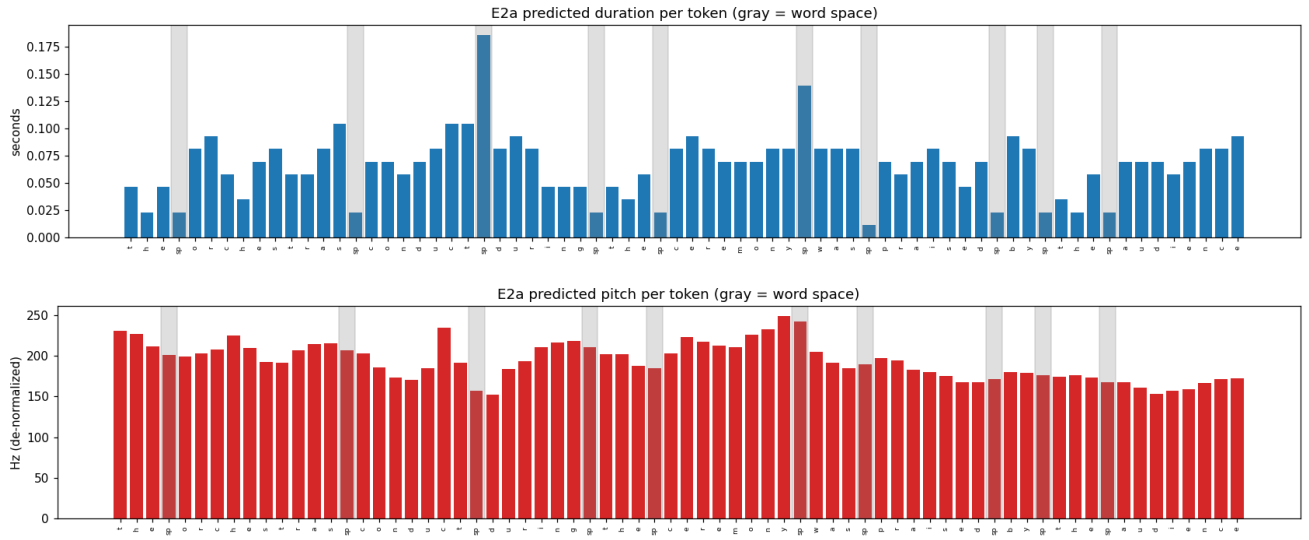


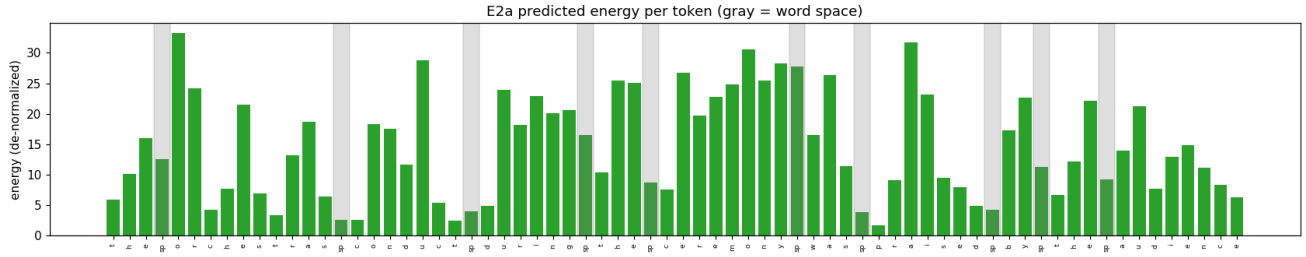
E2a

The orchestra's conduct during the ceremony was praised by the audience.

Speech observations. Whisper-small heard: “the orchestra's conduct during the ceremony was prized by the audience.”. WER 0.09 (case, punctuation, hyphens and apostrophes ignored).

Duration. 60 character tokens in 11 words; predicted length 4.64 s (rendered audio 4.64 s). Mean character duration 69 ms (max 104 ms); word spaces average 50 ms and take 10.8% of the time. Longest tokens: 's' (token 13, 104 ms), 't' (token 21, 104 ms), 'c' (token 20, 104 ms).
Pitch (F0). Mean 194 Hz, std 23 Hz, range 153–249 Hz. The first half of the sentence averages 202 Hz and the second half 187 Hz (a fall of 15 Hz). Highest pitch on 'y' (token 41).
Energy. Mean 15.6, range 1.7–33.2; peak on 'o' (token 4 of 70, 6% into the sentence). Duration–energy correlation across characters: +0.09.



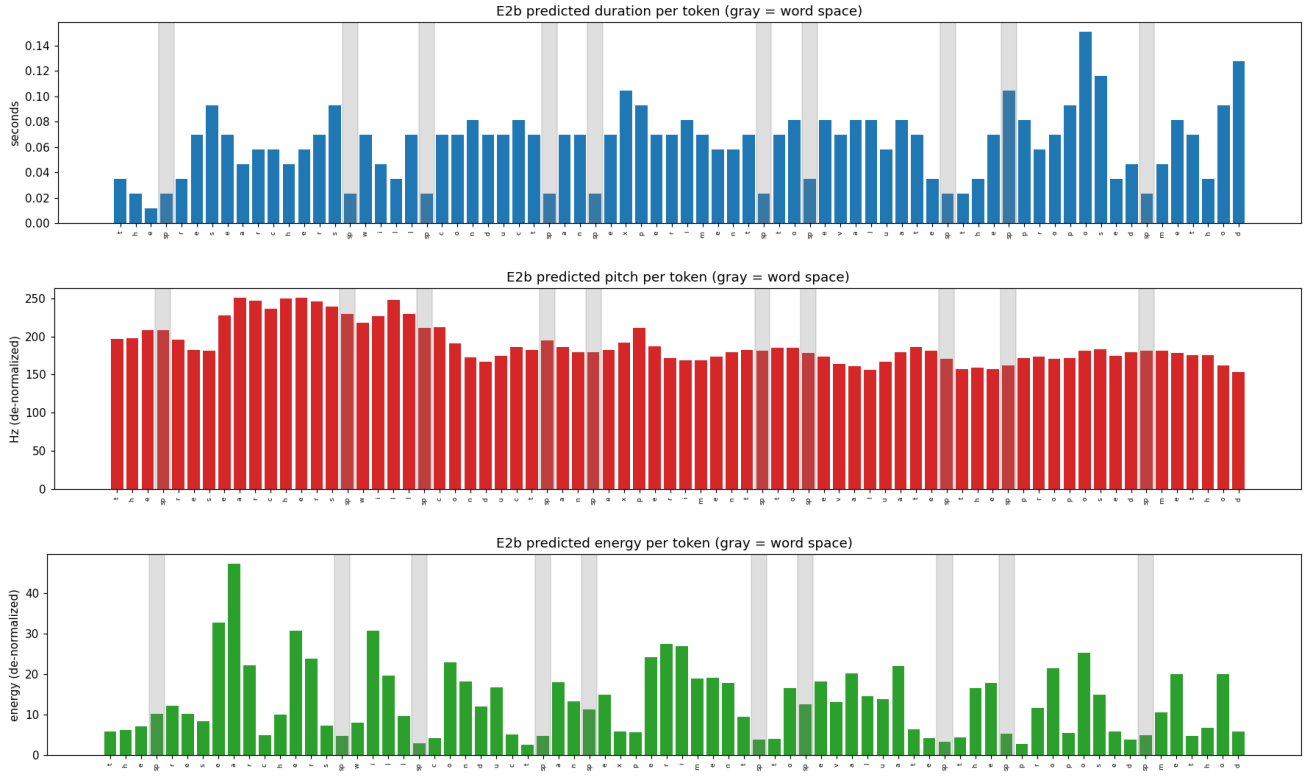


E2b

The researchers will conduct an experiment to evaluate the proposed method.

Speech observations. Whisper-small heard: “The searchers will conduct an experiment to evaluate the proposed method.”. WER 0.09 (case, punctuation, hyphens and apostrophes ignored).

Duration. 64 character tokens in 11 words; predicted length 4.64 s (rendered audio 4.64 s). Mean character duration 67 ms (max 151 ms); word spaces average 33 ms and take 7.0% of the time. Longest tokens: 'o' (token 63, 151 ms), 'd' (token 73, 128 ms), 's' (token 64, 116 ms).
Pitch (F0). Mean 190 Hz, std 27 Hz, range 153–251 Hz. The first half of the sentence averages 207 Hz and the second half 172 Hz (a fall of 35 Hz). Highest pitch on 'a' (token 8).
Energy. Mean 14.2, range 2.6–47.2; peak on 'a' (token 8 of 74, 11% into the sentence). Duration–energy correlation across characters: +0.08.



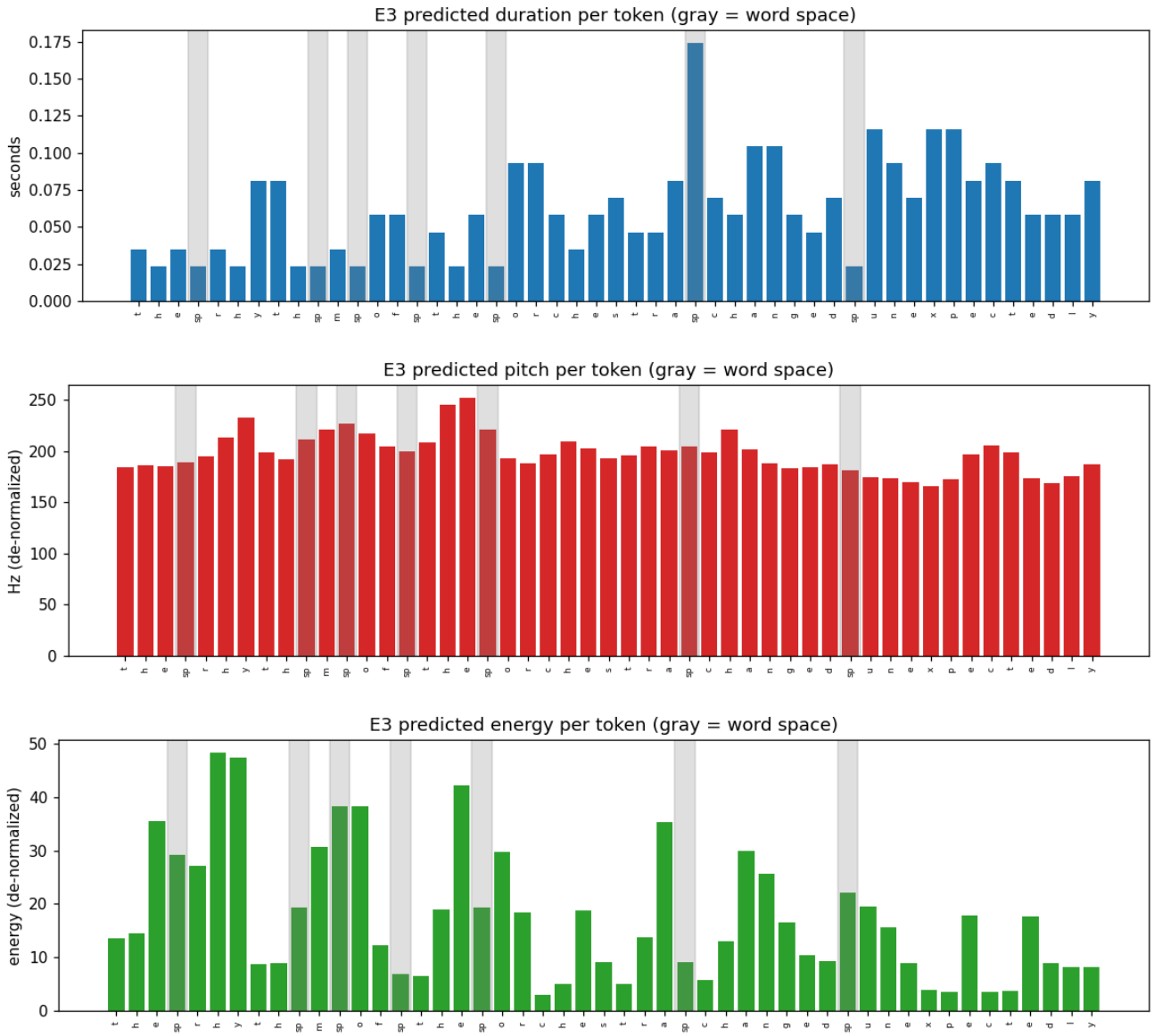
E3

The rhyth m of the orchestra changed unexpectedly.

Speech observations. Whisper-small heard: “The rate of the Orphestra changed unexpectedly.”. WER against the literal input (“rhyth m”): 0.38; against the intended phrase (“rhythm”): 0.29.

Duration. 42 character tokens in 8 words; predicted length 3.04 s (rendered audio 3.04 s). Mean character duration 65 ms (max 116 ms); word spaces average 45 ms and take 10.3% of the time. Longest tokens: 'x' (token 40, 116 ms), 'p' (token 41, 116 ms), 'u' (token 37, 116 ms).

Pitch (F0). Mean 196 Hz, std 19 Hz, range 166–252 Hz. The first half of the sentence averages 207 Hz and the second half 188 Hz (a fall of 19 Hz). Highest pitch on 'e' (token 17).
Energy. Mean 17.2, range 3.0–48.3; peak on 'h' (token 5 of 49, 10% into the sentence). Duration–energy correlation across characters: -0.11.



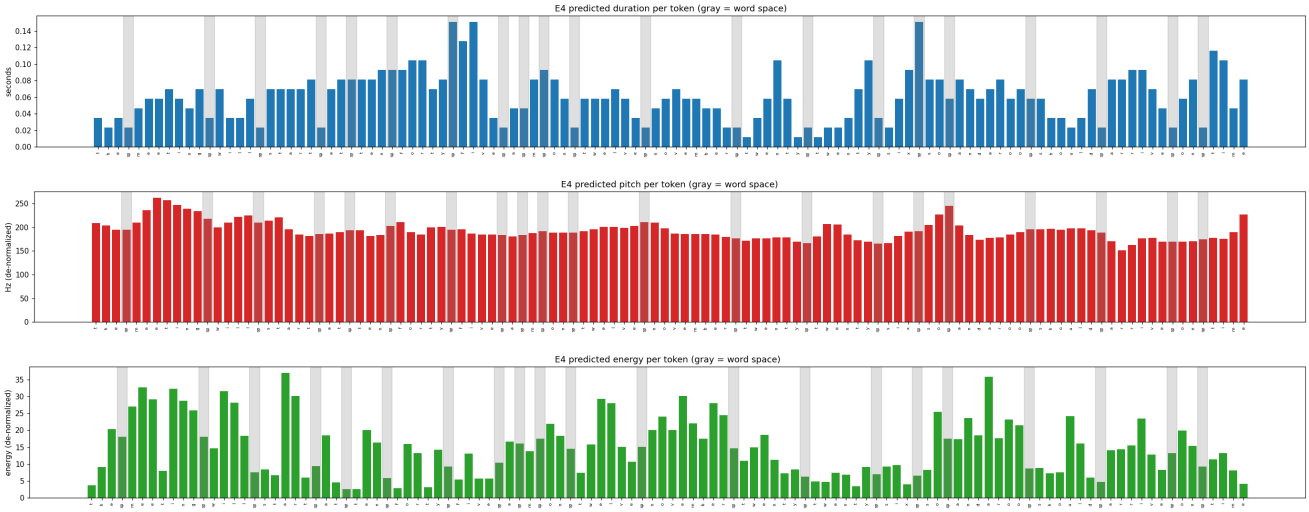
E4

The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.
synthesized as: The meeting will start at ten forty five a m on twelve november twenty twenty six, so andaroo should arrive on time.

Speech observations. Whisper-small heard: “The meeting will start at 10.45. I am on 12th November. I need to take it. So, and a row should I raise on time.”. WER not computed (ASR writes numerals, the input was spelled out).

Duration. 93 character tokens in 22 words; predicted length 6.95 s (rendered audio 6.95 s). Mean character duration 63 ms (max 151 ms); word spaces average 50 ms and take 15.2% of the time. Longest tokens: 'i' (token 37, 151 ms), 'f' (token 36, 128 ms), 't' (token 110, 116 ms).
Pitch (F0). Mean 194 Hz, std 20 Hz, range 151–262 Hz. The first half of the sentence averages 204 Hz and the second half 184 Hz (a fall of 20 Hz). Highest pitch on 'e' (token 6).

Energy. Mean 15.6, range 2.6–37.0; peak on 'a' (token 19 of 114, 17% into the sentence). Duration–energy correlation across characters: -0.07.



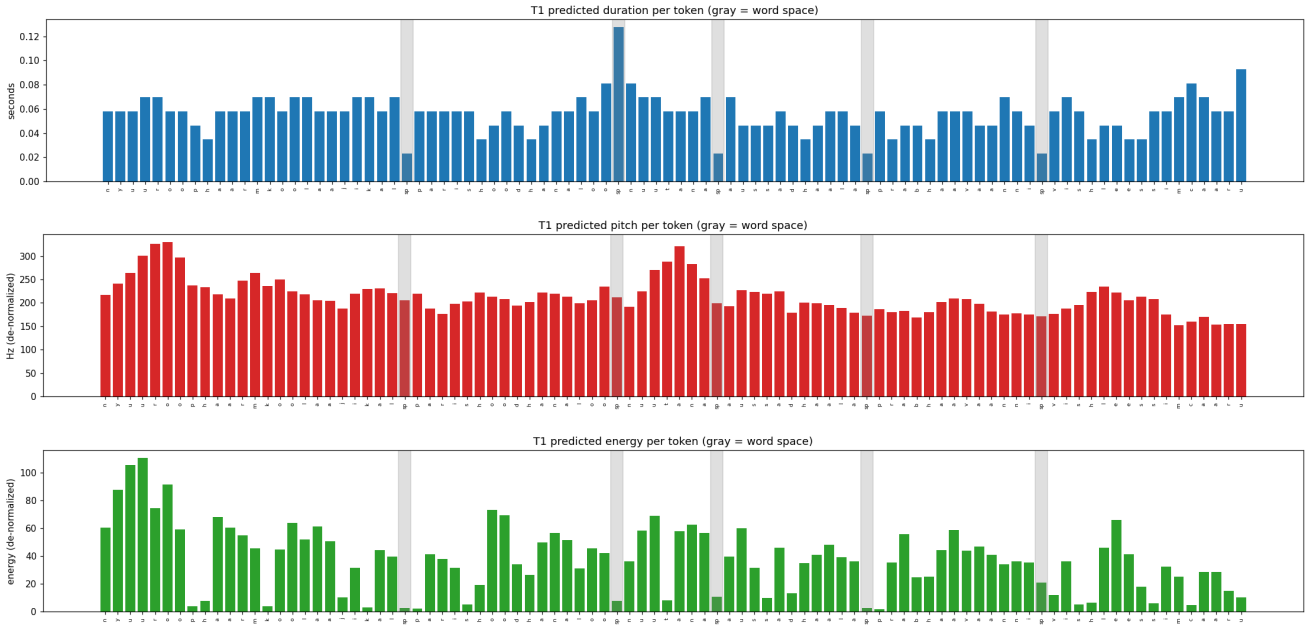
5.4 Telugu sentences

T1

న్యూరోఫార్మ్కాలజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

Speech observations. No ASR or human listening test was performed for Telugu; the observations below are objective diagnostics only (the model produced a complete waveform of plausible length, with no collapsed or empty tokens).

Duration. 87 character tokens in 6 words; predicted length 5.18 s (rendered audio 5.18 s). Mean character duration 57 ms (max 93 ms); word spaces average 44 ms and take 4.3% of the time. Longest tokens: 'u' (token 91, 93 ms), 'o' (token 40, 81 ms), 'c' (token 87, 81 ms).
Pitch (F0). Mean 214 Hz, std 38 Hz, range 152–330 Hz. The first half of the sentence averages 230 Hz and the second half 198 Hz (a fall of 32 Hz). Highest pitch on 'o' (token 5).
Energy. Mean 39.8, range 1.6–110.6; peak on 'u' (token 3 of 92, 3% into the sentence). Duration–energy correlation across characters: +0.11.

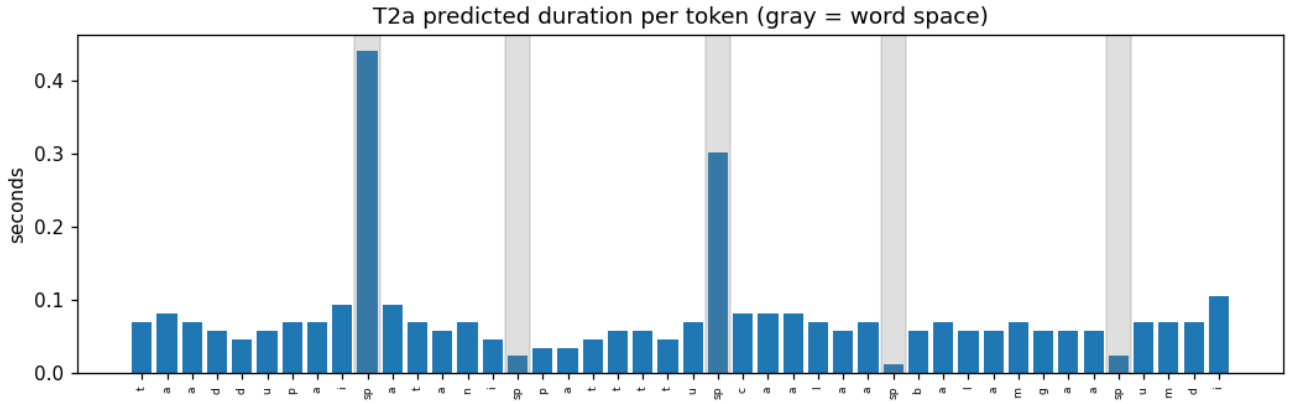


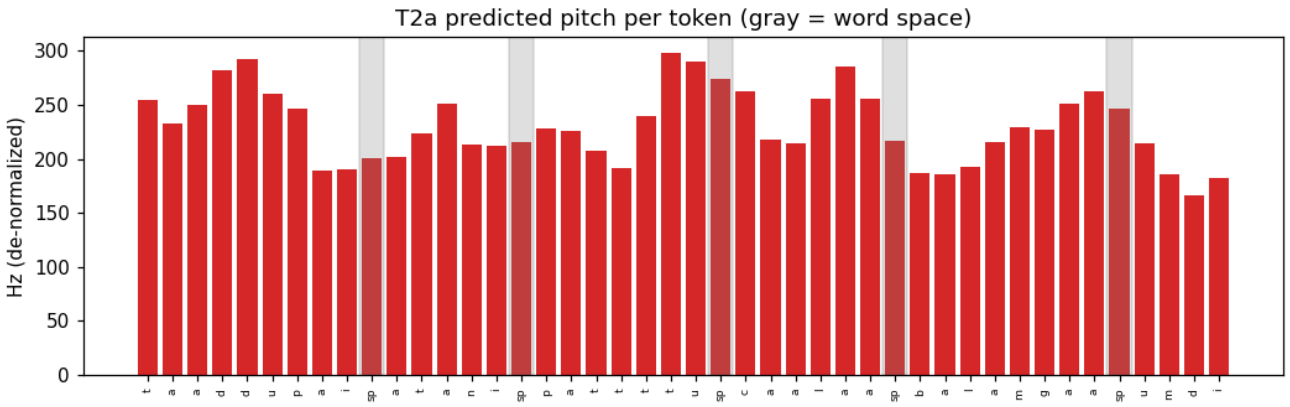
T2a

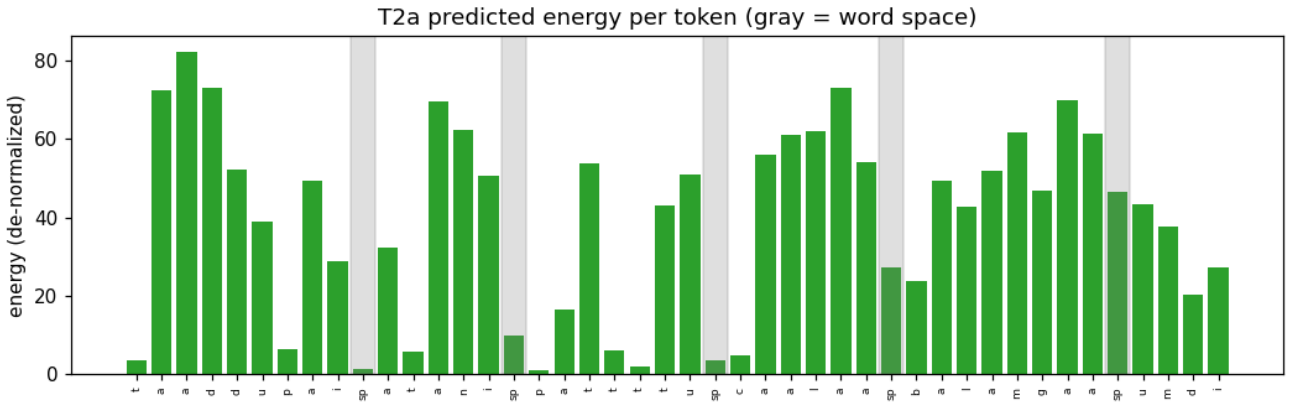
తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

Speech observations. No ASR or human listening test was performed for Telugu; the observations below are objective diagnostics only (the model produced a complete waveform of plausible length, with no collapsed or empty tokens).

Duration. 39 character tokens in 6 words; predicted length 3.34 s (rendered audio 3.34 s). Mean character duration 65 ms (max 104 ms); word spaces average 160 ms and take 24.0% of the time. Longest tokens: 'i' (token 43, 104 ms), 'a' (token 10, 93 ms), 'i' (token 8, 93 ms).
Pitch (F0). Mean 230 Hz, std 34 Hz, range 166–298 Hz. The first half of the sentence averages 234 Hz and the second half 225 Hz (a fall of 9 Hz). Highest pitch on 't' (token 21).
Energy. Mean 42.3, range 1.1–82.2; peak on 'a' (token 2 of 44, 5% into the sentence). Duration–energy correlation across characters: +0.02.





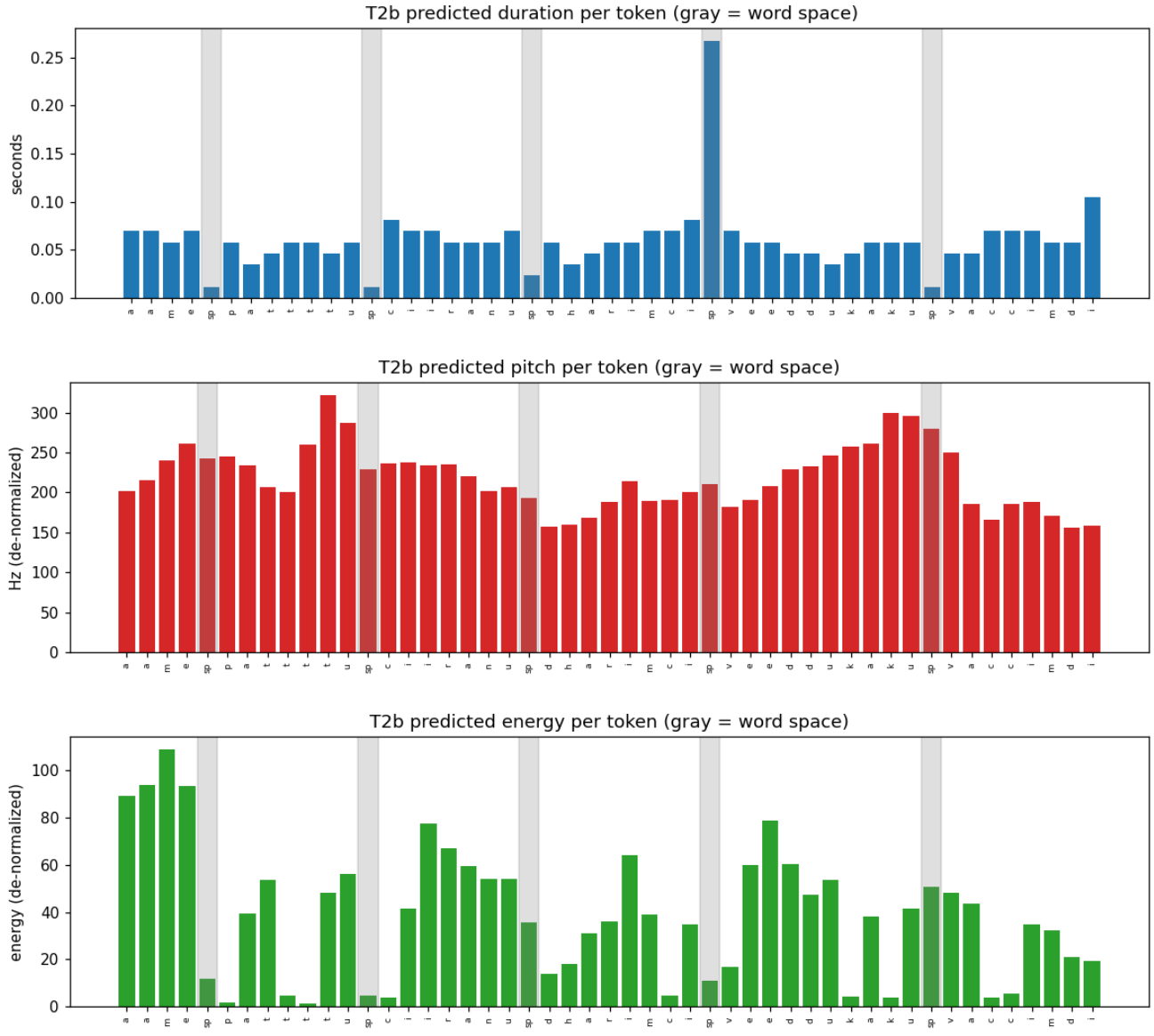


T2b

ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

Speech observations. No ASR or human listening test was performed for Telugu; the observations below are objective diagnostics only (the model produced a complete waveform of plausible length, with no collapsed or empty tokens).

Duration. 44 character tokens in 6 words; predicted length 2.95 s (rendered audio 2.95 s). Mean character duration 60 ms (max 104 ms); word spaces average 65 ms and take 11.0% of the time. Longest tokens: 'i' (token 48, 104 ms), 'c' (token 13, 81 ms), 'i' (token 28, 81 ms).
Pitch (F0). Mean 218 Hz, std 40 Hz, range 155–322 Hz. The first half of the sentence averages 225 Hz and the second half 211 Hz (a fall of 15 Hz). Highest pitch on 't' (token 10).
Energy. Mean 40.9, range 1.1–108.9; peak on 'm' (token 2 of 49, 4% into the sentence). Duration–energy correlation across characters: -0.07.

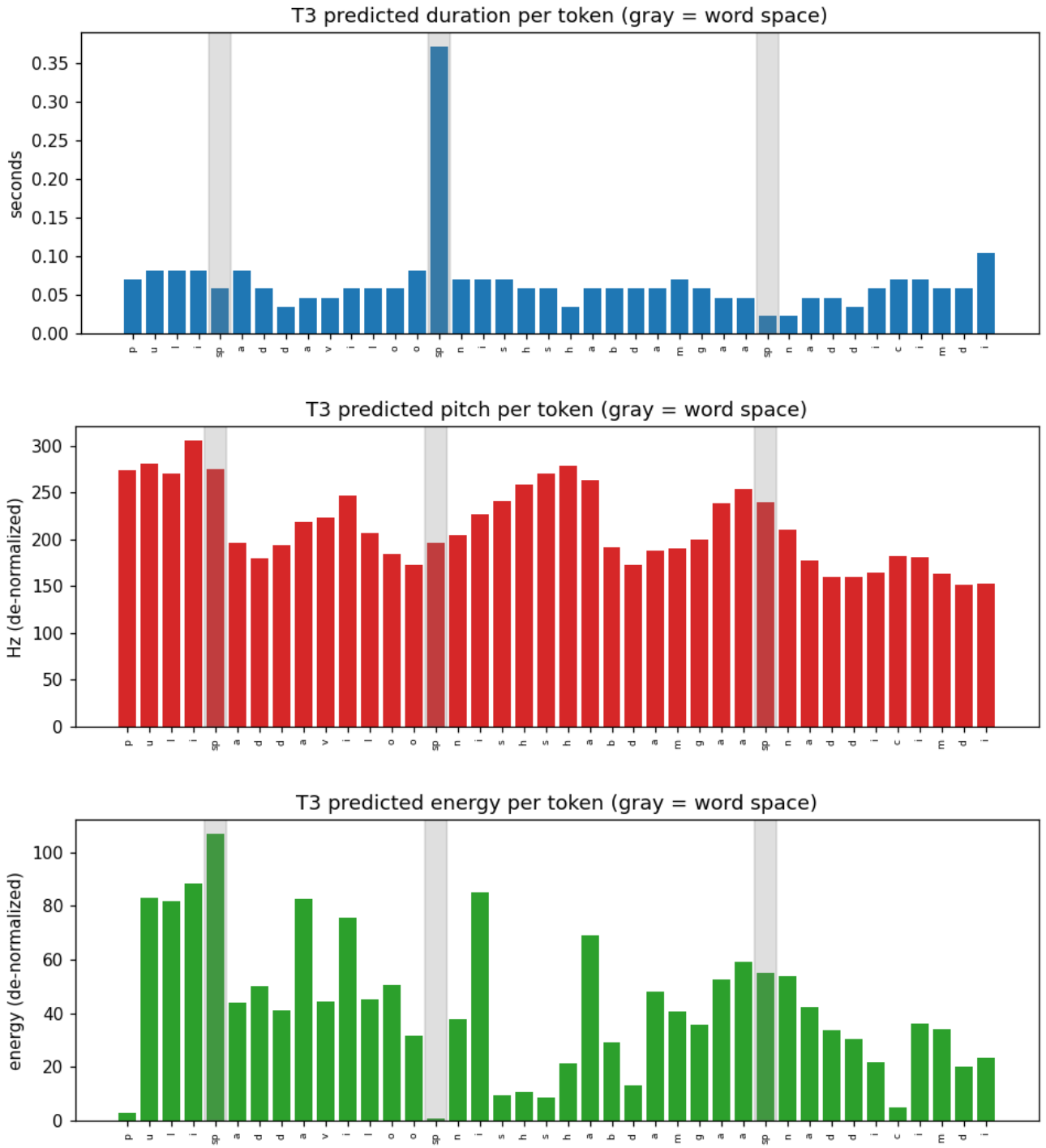


T3

పులి అడవిలో నిశ్శబ్దంగా నడిచింది.

Speech observations. No ASR or human listening test was performed for Telugu; the observations below are objective diagnostics only (the model produced a complete waveform of plausible length, with no collapsed or empty tokens).

Duration. 37 character tokens in 4 words; predicted length 2.67 s (rendered audio 2.67 s). Mean character duration 60 ms (max 104 ms); word spaces average 151 ms and take 17.0% of the time. Longest tokens: 'i' (token 39, 104 ms), 'u' (token 1, 81 ms), 'i' (token 3, 81 ms).
Pitch (F0). Mean 212 Hz, std 42 Hz, range 152–305 Hz. The first half of the sentence averages 231 Hz and the second half 194 Hz (a fall of 37 Hz). Highest pitch on 'i' (token 3).
Energy. Mean 41.7, range 3.0–88.3; peak on 'i' (token 3 of 40, 8% into the sentence). Duration–energy correlation across characters: +0.05.

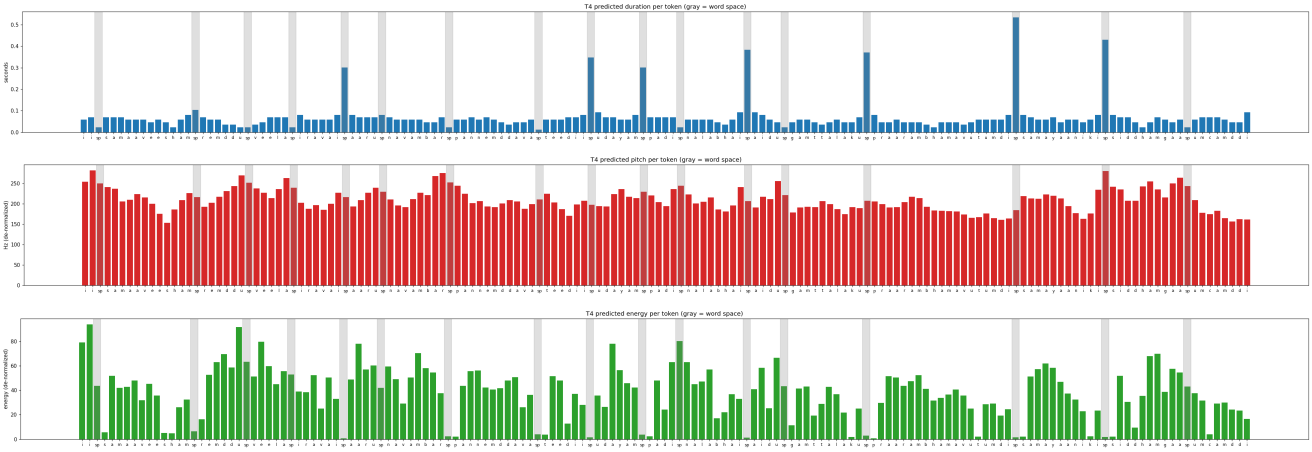


T4

ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.
synthesized as: ఈ సమావేశం రెండు వేల ఇరవై ఆరు నవంబర్ పన్నెండవ తేదీ ఉదయం పది నలభై ఐదు గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

Speech observations. No ASR or human listening test was performed for Telugu; the observations below are objective diagnostics only (the model produced a complete waveform of plausible length, with no collapsed or empty tokens).

Duration. 140 character tokens in 18 words; predicted length 11.26 s (rendered audio 11.26 s). Mean character duration 59 ms (max 93 ms); word spaces average 178 ms and take 26.9% of the time. Longest tokens: 'a' (token 90, 93 ms), 'i' (token 88, 93 ms), 'u' (token 69, 93 ms).
Pitch (F0). Mean 207 Hz, std 26 Hz, range 153–282 Hz. The first half of the sentence averages 214 Hz and the second half 199 Hz (a fall of 15 Hz). Highest pitch on 'i' (token 1).
Energy. Mean 39.6, range 0.7–93.7; peak on 'i' (token 1 of 157, 1% into the sentence). Duration–energy correlation across characters: -0.08.



6. How the variance adaptor shapes the output

Duration. The predictor outputs a log-duration per token; the length regulator repeats each token's hidden state that many mel frames, so the predicted durations directly set the length of the spectrogram and the audio (rendered audio length equals the summed predicted duration in every test sentence). Predicted character durations are tightly clustered (about 58–71 ms means) and are multiples of one mel frame (11.6 ms); word-space tokens vary most between sentences. **Pitch and energy.** Each predicted value is quantised into one of 256 bins and its embedding is added to the encoder output before the decoder, so these predictions shape the mel contour (pitch harmonics, loudness) of each token. Duration and energy are uncorrelated across characters (|r| mostly near 0), consistent with independent predictors on the same encoder output.

7. English vs Telugu

- **Pitch level and range.** Mean predicted F0 is about 193 Hz (English, mean within-sentence std 22 Hz) vs 216 Hz (Telugu, std 36 Hz). The models were trained on different speakers, so this is not evidence of a language effect. Telugu pitch moves over a wider range; the English contours are flatter, which fits the known tendency of FastSpeech 2 predictors toward mean values (not tested here).
- **Declination.** Every sentence in both languages has a lower second-half pitch than first-half, as expected for declarative sentences.
- **Pauses.** Word spaces take 11% of predicted time in English and 17% in Telugu on average, with large sentence-to-sentence variation (for example T2a vs T2b).

- **Energy.** Peak energy sits on average 11% (English) and 4% (Telugu) of the way into a sentence; the Telugu peaks fall within the first four tokens in all five sentences, an onset-emphasis pattern. For E4 the English peak is near the end of the sentence.
- **Overall English quality (Whisper-small proxy, no human test).** E2a and E2b were transcribed almost correctly (one word substituted each: “prized” for “praised”, “searchers” for “researchers”). E1’s long word was split and misrecognised and “gained” was heard as “is banned”. E3 was garbled (“rate of the Orphestra”). In E4 the time and month were recognised, but the rest of the sentence and “andaroo” were not. English output is therefore intelligible on ordinary sentences and unreliable on rare words, odd spellings and numeral clauses.
- **Overall Telugu quality.** No ASR or listening test was available, so Telugu intelligibility is *not* verified. The diagnostics show complete, plausible-length outputs with sensible pitch declination and no collapsed tokens, which is necessary but not sufficient for good speech.
- **Failures.** E4 and T4 contain the date/time clause, and E4 also the out-of-vocabulary “andaroo”. Because the models see spelled-out characters with no lexicon, rare long words (“antidisestablishmentarianism”, the Telugu technical word in T1) and number expansions are the likely failure points; English ASR transcripts are reported per sentence in Section 5.3.

8. Limitations and possible improvements

- Both datasets are about 15% below the 5 h / 1 h target; 328 English clips were unaligned and 711 English alignments came from a heuristic recovery that was only spot-checked.
- Telugu has two proxy speakers and a clip-level split (not speaker-independent); English validation speakers all appear in training.
- No human evaluation; English quality is judged by Whisper-small, whose scores were noisy, and Telugu has no ASR check.
- English validation loss plateaued early and the duration/pitch losses drift up, so earlier stopping or regularisation of the variance predictors could help; a model saved at step 7,000 does not exist.
- Improvements: a text normaliser for numbers and dates, G2P or a pronunciation lexicon for English, punctuation tokens for pauses, more data, and speaker-held-out validation.

9. Reproducibility

Environment. Python 3.13, PyTorch 2.11 (CUDA 12.8; install the CUDA build first: `pip install torch --index-url https://download.pytorch.org/whl/cu128`), then `pip install -r requirements.txt` (includes ctc-forced-aligner 1.0.2, uroman, onnxruntime-gpu). Trained on Kaggle, 2× T4.

```
cd English   # or Telugu
# 1. data: write wav + .lab to raw_data/<Lang>/<speaker>/ (clip lists in data_prep/)
python alignment/align_english.py 0 5000      # Telugu: alignment/align_telugu.py 0 5000
python alignment/recover_english.py           # English only: retry rejected clips
python data_prep/rebuild_split.py             # English: merge, features, rebuild train/val
python preprocess.py config/Telugu/preprocess.yaml   # Telugu feature extraction
# 2. train
python train.py -p config/<Lang>/preprocess.yaml -m config/<Lang>/model.yaml -t config/<Lang>/
train.yaml
# 3. inference (checkpoint at output/ckpt/<Lang>/20000.pth.tar, vocoder at hifigan/
generator_universal.pth.tar)
python inference.py --restore_step 20000 --speaker_id 94 --sentences test_sentences.json --out
out   # Telugu: --speaker_id 0
```

Checkpoints and vocoder: https://huggingface.co/rohan1822/fastspeech2-tts-checkpoints (English/20000.pth.tar, Telugu/20000.pth.tar, vocoder/generator_universal.pth.tar).

References

Y. Ren et al., FastSpeech 2: Fast and High-Quality End-to-End Text to Speech, ICLR 2021. · ming024/FastSpeech2 (GitHub). · J. Kong et al., HiFi-GAN, NeurIPS 2020. · Svarah (AI4Bharat). · IndicTTS Telugu (SPRINGLab). · ctc-forced-aligner / MMS (Meta).